

WEEK 6 IMPLEMENTATION REPORT

Assessment & Quiz Engine — Data Model, Transactional Authoring, Server-Side Grading & Progress Analytics

Project Phase:	Week 6 Implementation	Status:	COMPLETED (70/70 Tests Passed)
Author:	Backend Architecture Team	Date:	September 2, 2026
Technology Stack:	Spring Boot, Java 21, JPA, PostgreSQL, JWT	Test Suite:	39 Assessment + 31 Base = 70 Tests

1. Executive Summary & Objective

Week 6 successfully delivers the core **Assessment and Quiz Engine** and **Single-Page Scrollable Quiz UI** for LearnPulse LMS. The system allows Teachers and Admins to author structured multiple-choice quizzes atomically with nested questions, enables Students to discover and attempt quizzes via a Google Forms-style vertical layout, supports option selection with invisible correctness feedback, provides controlled single-question answer reveals displaying actual correct answers, and performs strict server-side grading to calculate score, percentage, and detailed answer metrics. Correct answers are kept secure and are **never exposed** to students in initial quiz payloads or list DTOs.

2. System Architecture & Component Design

The engine follows Spring Boot layered architecture: Single Page UI -> Controller -> Service -> Repository -> PostgreSQL DB.

```
Web Quiz UI (/quiz.html) -> TeacherQuizController / StudentQuizController
--> QuizService (Authoring, Delivery & Controlled Reveal)
--> QuizGradingService (Server-Side Autonomous Grading)
--> QuizRepository / StudentQuizResultRepository --> PostgreSQL
```

3. Relational Schema & JPA Entities

Entity	Table Name	Key Fields & Relations	Cascade / Constraints
Quiz	quizzes	id, title, description, subject_id, chapter_id, total_marks, created_by	OneToMany(questions), Cascade.ALL, orphanRemoval
Question	questions	id, quiz_id, question_text, option_a..d, correct_answer, marks	ManyToOne(quiz), NOT NULL, length=1
StudentQuizResult	student_quiz_results	id, student_id, quiz_id, score, total_marks, percentage, correct/wrong	ManyToOne(student, quiz), timestamped

4. Endpoints & RESTful API Specifications

HTTP Method & Route	Required Role	Description & Security Guarantee
POST /api/teacher/create-quiz	TEACHER, ADMIN	Atomic creation of quiz and nested questions.
GET /api/quizzes	AUTHENTICATED	List quizzes with optional subject/chapter filter. Strips correctAnswer.
GET /api/quizzes/{id}	AUTHENTICATED	Fetch quiz delivery details. Strips correctAnswer.
POST .../questions/.../reveal	AUTHENTICATED	Single-question reveal. Validates prerequisite option selection.
POST /api/quizzes/submit	AUTHENTICATED	Submit student choices. Backend performs 100% of grading.
GET /api/quizzes/result/{id}	AUTHENTICATED	Get result attempt. Enforces student ownership.
GET /api/student/progress	STUDENT	Aggregated analytics (avg score, %, attempts).

5. Automated Test Suite Execution Results

The automated test suite runs **39 comprehensive integration tests** in `AssessmentEngineIntegrationTest` covering all required test cases (quiz authoring, validation failures, student delivery without correct answers, controlled single-question answer reveals with prerequisite validation, scoring independence from answer reveals, result persistence, student ownership security, repeated attempt tracking, and performance analytics).

Test Class Name	Tests Run	Pass / Fail	Coverage Area
<code>AssessmentEngineIntegrationTest</code>	39	39 PASSED	Quiz authoring, grading, reveal security & progress
<code>DocumentIngestionIntegrationTest</code>	9	9 PASSED	PDF/DOC/DOCX file ingestion & extraction
<code>SecurityRbacIntegrationTest</code>	7	7 PASSED	Auth, JWT, RBAC & privilege escalation
<code>AcademicHierarchyIntegrationTest</code>	5	5 PASSED	Subject & Chapter CRUD
<code>StudentTeacherProfileIntegrationTest</code>	5	5 PASSED	Profile management & ownership
<code>JwtProviderTest</code> & Base Status	5	5 PASSED	JWT token generation & status checks
TOTAL PROJECT TEST SUITE	70	70 PASSED (0 FAIL)	BUILD SUCCESS (35.9s execution time)

6. Verification & Mentor Sign-Off

Verification Confirmed: All Week 6 Assessment and Quiz Engine requirements have been implemented, persisted in PostgreSQL, and verified with 66 passing automated tests. Correct answers are strictly hidden from student responses, server-side grading is 100% autonomous, and security ownership is enforced.